

Lab Report: Magnetic Field of a Solenoid

Chhaya Ramnani (UID: u8336188)

Lab partners: Tsz Wing Candice Ng, Chinatsu Komada

2026-08-19

Imports and Functions

Pre-lab Question

1. For N loops connected in series, each loop develops the same emf $\varepsilon_1 = -d\Phi/dt$, and the total is just the sum over all loops:

$$\varepsilon = -N \frac{d\Phi}{dt}$$

so we expect a linear dependence on N .

2. The flux through one loop is $\Phi = B_{\perp} A = B_{\perp} \pi R^2$. Since R only enters through the area, so for a fixed magnet/trajectory we expect $\text{emf} \propto 1/R^2$ dependence.
3. For the speed dependence, use the chain rule on position z :

$$\varepsilon = -N \frac{d\Phi}{dt} = -N \frac{d\Phi}{dz} \frac{dz}{dt} = -N s \frac{d\Phi}{dz}$$

For a fixed drop $d\Phi/dz$ is a fixed function of the magnet's position relative to the coil, so the peak emf should scale **linearly** with s .

Aim

Theory

Procedure

Results

Observations